Q222 Analytical Extraction Refinement

Regenerated proofs, exact coefficient data, and the remaining global gates

Mathematical reconstruction and verification record

22 October 2026

Abstract

This release reconstructs the analytical extraction refinement from retained sources and newly generated exact data. It delivers an evaluated local complete-parent regular-C2 certificate, an exact affine pencil for the tilted–CHSH two-coordinate projection, the displayed B_0 coefficient module, restricted contact data, and the general local constructions used to form fixed features. It also proves a precise limitation of literal Schur-remainder extraction and specifies a conditional exact algebraic search for a full lift. The global feature list, full eight-coordinate affine pencil, and both global inclusions remain open. The accompanying archive contains the complete sources and replayable coefficient checks, as well as the earlier continuation archive unchanged.

1 Outcome, scope, and source custody

Write the original quantum coordinates as

$$q = (a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}) \in \mathbb{R}^8.$$

The purpose of exact finite-lift extraction is to evaluate finite integers N, m and all real symmetric matrices in

$$\mathcal{Q}_{222} = \left\{ q \in \mathbb{R}^8 : \exists y \in \mathbb{R}^m, A_0 + \sum_{\nu=1}^{8} q_\nu A_\nu + \sum_{j=1}^{m} y_j B_j \succeq 0 \right\}, \tag{1}$$

and to prove both inclusions for those same coefficients. Equation (1) is the target of the open global extraction items, not a newly evaluated representation delivered by this report.

The regenerated results have the following precise domains.

Construction	Delivered data	Exact scope
Regular C2	785 blocks of order two; 237 ambient product monomials	One normalized contact graph and an independent testing box.
Tilted–CHSH	Order five, one auxiliary; or order four, no auxiliary	The projection $(a_0, E_{00} + E_{01} + E_{10} - E_{11})$.
Displayed B_0	Nine square features; order-four diagonal coefficient pencil	The four-coefficient module, with original factor typing preserved.
Smooth diagonal	Explicit finite local separation and frame construction	Exact J^2 data and a positive tangential residue are hypotheses.
Global search	Rational physical formula and exact unsolved QE input	A conditional decision procedure; no global coefficient tuple.

The regular-C2 certificate keeps the complete polynomial parent and separates source and testing variables. Its local feature list and product map are fully evaluated. The tilted construction additionally supplies an explicit matrix pencil and a proof of both inclusions for its projection. A rational nonquantum behavior shows why this family, even after relabelings and probability constraints, does not describe the full set. The global finite-cover theorem states exactly what further data would turn sufficiently many restored local certificates into (1).

What was recovered. The canonical August 30 source has 76,839 bytes and SHA-256

`e268a6ae1c11dd02b5819064fa7e8187173074050d92f3204e703dd46f574731`.

It was recovered from the retained continuation archive through its nested preceding releases. The attached August 31 frozen source archive has 17,609 bytes and SHA-256

`1a6208d7d1f51e5e166d46e80be3167ffc43aff97027f5d0c39705ab5d4f2bb3`.

The earlier continuation archive is preserved unchanged at 8,884,930 bytes, with SHA-256

`9dcf9554b1a2578bb6b9fd909e7d194d82636fe61a0035da0694eb2e44f441a3`.

The file `inputs/SOURCE_CUSTODY.json` records the recovery chain. Matching hashes establish that these inputs have been preserved. They do not revalidate every inherited mathematical claim.

What this regeneration does not restore. The historical 549-page R21 proof and its original completeness assertion are not recreated or independently recertified here. This is a newly regenerated refinement with its own checked data, not a byte-for-byte copy of a lost report. The unchanged prior archive retains the preceding derivations and candidate matrices for continued work. No earlier release has been overwritten.

How to read and reproduce the release. The next section gives the complete local C2 argument and evaluated compiler conventions. The local separation, B_0, contact, and assembly sections explain how those methods extend and what is still missing. The tilted section gives the fully evaluated projected example. The Schur-remainder and effective-search sections delimit two other extraction routes. The final checklist records all twenty supplementary acceptance items and ten concrete global data obligations. Run `python3 verify.py` for integrity, or `python3 verify.py -replay quick` for the fresh exact suites in an isolated temporary copy. The root verification report states the scope and outcome of the actual release checks.

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2 Complete-parent regular-C2 local extraction

2.1 Complete parent and the precise domain

The seven source coordinates, in their fixed order, are

$$(d, n, m, C, A, D, \alpha).$$

Write $D_u = 1 + u^2$, $D_v = 1 + v^2$. The complete homogeneous quarter-parent is

$$\begin{aligned}\widetilde{P} = {} & nD_uD_vX^2 + m(vW - X + uvY - uZ)^2 \\ & + CD_v(uW - Y)^2 + AD_uD_vY^2 \\ & + (D + d/2)(uW + uvX - Y - vZ)^2 \\ & + (\alpha + d/2)D_u(vY - Z)^2 + (d/2)D_v(X + uZ)^2 - (d/2)D_uD_vZ^2. \end{aligned} \tag{2}$$

Every term in (2) is retained. For the local certificate set $d = Z = 1$, let $c = (n, m, C, A, D, \alpha)$ and $z = (u, v, W, X, Y)$, and denote the resulting polynomial by $P_c(z)$. The testing-variable order is fixed throughout.

The source/contact anchor is

$$c_* = (7/8, 9/8, 1/18, 11/8, 9/8, 1/18), \qquad \theta_* = (1, 1, 1, -1/10, 1/10). \tag{3}$$

For $\theta = (p, q, a, b, r)$ define the six contact equations

$$F(c, \theta) = \big(P_c(\theta), \partial_u P_c(\theta), \partial_v P_c(\theta), \partial_W P_c(\theta), \partial_X P_c(\theta), \partial_Y P_c(\theta)\big) = 0. \tag{4}$$

These equations impose the complete value and gradient conditions. In the original homogeneous state variables, the value equation and the three displayed state derivatives also imply the missing Z derivative by Euler's identity for a homogeneous quadratic. No state-kernel condition is discarded.

Let the eleven centered source/contact variables and five independently centered testing variables be

$$s = (c - c_*, h), \qquad h = \theta - \theta_*, \qquad t = z - \theta_*. \tag{5}$$

Their order is the six coordinates of c, followed by the five coordinates of h; the order of t is that of z. Set

$$\rho = \frac{510101005}{350806159660744}. \tag{6}$$

The source domain used in this record is exactly

$$\mathcal{S}_\rho = \{(c, \theta) : F(c, \theta) = 0, |s_i| \le \rho \ (1 \le i \le 11)\}, \qquad \mathcal{T}_\rho = \{t : |t_i| \le \rho \ (1 \le i \le 5)\}. \tag{7}$$

The source/contact point and the testing point vary independently. Stationarity alone is not asserted to imply global physical validity of a source.

Theorem 2.1 (Evaluated local certificate)**.** *For each $(c, \theta) \in \mathcal{S}_\rho$ there are 785 positive semidefinite matrices $Q_b(c, \theta) \in \mathbb{S}^2_+$. There are 785 fixed two-component testing feature vectors $\psi_b(t)$, independent of (c, θ), such that*

$$P_c(\theta_* + t) = \sum_{b=0}^{784} \psi_b(t)^T Q_b(c, \theta) \psi_b(t) \qquad (t \in \mathcal{T}_\rho). \tag{8}$$

All radicands, multipliers, and source Gram factors are explicitly evaluated in the accompanying JSON files. The symmetric products of these features are polynomials in the 237 listed monomials, of degree at most six. The complete seven-source parent has the coefficient matching map stated in Section 2.7.

2.2 The contact graph is regular

Substitution in (4) gives $F(c_*, \theta_*) = 0$. The derivative with respect to the six coordinates of c, at the anchor, is

$$J_* = \begin{pmatrix} 1/25 & 1/25 & 81/50 & 1/25 & 1/25 & 81/50 \\ 1/25 & -9/25 & 18/5 & 1/25 & -9/25 & 81/50 \\ 1/25 & 11/25 & 81/50 & 1/25 & 11/25 & -9/25 \\ 0 & 2/5 & 18/5 & 0 & -2/5 & 0 \\ -4/5 & -2/5 & 0 & 0 & -2/5 & 0 \\ 0 & 2/5 & -18/5 & 4/5 & 2/5 & -18/5 \end{pmatrix}, \qquad \det J_* = \frac{114048}{78125} \neq 0. \tag{9}$$

Thus the contact equations define a five-dimensional graph near the anchor. There is also a direct uniform regularity check on the declared contact box. Since F is affine in c, its source Jacobian $J(\theta)$ depends only on θ. Put $E(h) = J_*^{-1} J(\theta_* + h) - I_6$. Its entries are rational polynomials with zero constant term. Direct exact coefficient collection gives

$$\max_i \sum_j \sum_{\beta \neq 0} |[h^\beta] E_{ij}(h)| = \frac{114199}{160}. \tag{10}$$

Consequently, for $|h_i| \leq \rho \leq 1$,

$$\|E(h)\|_\infty \leq \frac{11650604933999}{11225797109143808} < 1. \tag{11}$$

If $(I + E)x = 0$ and $x \neq 0$, choosing a component of x of maximal absolute value contradicts (11). Hence $J(\theta)$ is invertible on the whole contact box. Equivalently, the source coordinates are the unique rational solution of the six affine contact equations for each fixed contact point in that box; the definition of $\mathcal{S}_\rho$ further retains the source-coordinate bounds. No global property of that rational graph follows from this local statement.

2.3 Exact polynomial Taylor matrix

Expand $P_c(\theta + \Delta)$ in $\Delta = (\Delta_1, \ldots, \Delta_5)$, retaining polynomial coefficients in (c, θ). Remove the terms of Δ degree zero and one. For each remaining monomial apply the following deterministic allocation rule. If an exponent is at least two, use the first such coordinate and divide out its square, adding the result to the corresponding diagonal entry. Otherwise use the first two occupied coordinates and add half of the divided monomial to each of the two symmetric off-diagonal entries. This defines a symmetric polynomial matrix $H(c, \theta, \Delta)$ and proves the exact identity

$$P_c(\theta + \Delta) = P_c(\theta) + \nabla P_c(\theta)^T \Delta + \Delta^T H(c, \theta, \Delta) \Delta. \tag{12}$$

There is no asymptotic remainder in (12). All fifteen upper entries, both before and after substituting $\Delta = t - h$, are supplied as complete sparse rational polynomials.

At the anchor the matrix is half the Hessian:

$$H_0 = \begin{pmatrix} 10201/3600 & -1089/400 & 121/360 & 14/5 & -253/90 \\ -1089/400 & 10201/3600 & -9/40 & -14/5 & 253/90 \\ 121/360 & -9/40 & 103/36 & 1/2 & -11/18 \\ 14/5 & -14/5 & 1/2 & 29/4 & -11/4 \\ -253/90 & 253/90 & -11/18 & -11/4 & 341/36 \end{pmatrix}. \tag{13}$$

Its exact factorization $H_0 = L D_0 L^T$ is

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -9801/10201 & 1 & 0 & 0 & 0 \\ 1210/10201 & 8991/20002 & 1 & 0 & 0 \\ 10080/10201 & -5040/10001 & 79/1010 & 1 & 0 \\ -10120/10201 & 5060/10001 & -119/1010 & 2/81 & 1 \end{pmatrix}, \tag{14}$$

$$D_0 = \operatorname{diag}\left(\frac{10201}{3600}, \frac{20002}{91809}, \frac{27775}{10001}, \frac{891}{202}, \frac{1067}{162}\right) > 0. \tag{15}$$

In particular H_0 is positive definite. Define the rational linear functions $f = L^T z$, and put $\widehat{H} = L^{-1} H L^{-T}$. This orientation is essential: $\Delta f = L^T \Delta$ and

$$\Delta^T H \Delta = (\Delta f)^T \widehat{H} (\Delta f). \tag{16}$$

2.4 Twenty-five fixed linear frames

Set

$$\varepsilon = \frac{1}{20} \min_i (D_0)_{ii} = \frac{10001}{918090}. \tag{17}$$

Use the five frame functions f_i and, for every pair $i < j$, the two frame functions $f_i + f_j$ and $f_i - f_j$. Their respective coefficient polynomials are

$$\widehat{H}_{ii} - 8\varepsilon, \qquad \varepsilon + \tfrac{1}{2}\widehat{H}_{ij}, \qquad \varepsilon - \tfrac{1}{2}\widehat{H}_{ij}. \tag{18}$$

Order the five individual functions first; then order the pairs lexicographically, with the plus function preceding the minus function. The exact frame identity is

$$\begin{aligned} \Delta^T H \Delta = &\sum_i (\widehat{H}_{ii} - 8\varepsilon)(\Delta f_i)^2 \\ &+ \sum_{i<j} \left(\varepsilon + \tfrac{1}{2}\widehat{H}_{ij}\right)(\Delta f_i + \Delta f_j)^2 \\ &+ \sum_{i<j} \left(\varepsilon - \tfrac{1}{2}\widehat{H}_{ij}\right)(\Delta f_i - \Delta f_j)^2. \end{aligned} \tag{19}$$

Indeed each pair adds 2ε to each of its two diagonal entries, and its cross coefficient is $2\widehat{H}_{ij}$. Every coordinate participates in four pairs, giving the subtraction 8ε.

After inserting $c = c_* + s_{1:6}$, $\theta = \theta_* + h$, and $\Delta = t - h$, denote the 25 coefficient polynomials by $\gamma_k(s,t)$ and the 25 linear testing functions by $\ell_k(t)$. Their constant terms satisfy

$$\gamma_k(0,0) \geq \varepsilon. \tag{20}$$

For the ten plus and ten minus functions the constant is ε; for the five individual functions it is $(D_0)_{ii} - 8\varepsilon \geq 12\varepsilon$.

For a rational polynomial let its nonconstant coefficient norm mean the sum of the absolute values of all its nonconstant coefficients, with every source and testing monomial distinct. Exact coefficient collection gives

$$B_* := \max_k \sum_{(a,b)\neq(0,0)} |[s^a t^b]\gamma_k| = \frac{43803942777143}{46827180450}. \tag{21}$$

The radius in (6) is exactly

$$\rho = \min\left(\frac{1}{2}, \frac{\varepsilon}{8(1+B_*)}\right). \tag{22}$$

In particular $\rho B_* < \varepsilon/8$. The radius is also smaller than half the absolute value of each nonzero anchor coordinate in (3).

2.5 Positive separation of the two variable groups

Fix one frame and omit its index. Partition its complete coefficient support uniquely as

$$\gamma(s,t) = \gamma_0 + G(s) + T(t) + \sum_{\beta\neq 0} A_\beta(s) t^\beta, \tag{23}$$

where G, T, and every A_β have zero constant term. The sum ranges only over the nonzero mixed groups. Define

$$\sigma_\beta = \sum_a |[s^a]A_\beta|, \quad U_\beta(s) = A_\beta(s) + \sigma_\beta\rho, \quad V_\beta(t) = t^\beta + \rho^{|\beta|}, \quad B = \sum_\beta \sigma_\beta \rho^{|\beta|+1}. \tag{24}$$

The two pure remainders are the following explicit polynomials:

$$\begin{aligned} p(s) &= \gamma_0/2 - B/2 + G(s) - \sum_\beta \rho^{|\beta|} A_\beta(s), \\ q(t) &= \gamma_0/2 - B/2 + T(t) - \sum_\beta \sigma_\beta \rho\, t^\beta. \end{aligned} \tag{25}$$

Expanding every product proves, coefficient by coefficient,

$$\gamma(s,t) = p(s) + q(t) + \sum_\beta U_\beta(s) V_\beta(t). \tag{26}$$

The constant B is subtracted exactly once through the two halves $B/2$.

Lemma 2.2 (Uniform sign certificates)**.** *On the independent boxes in (7), all U_β and V_β are nonnegative, and for every frame*

$$p(s), q(t) \geq \frac{5\varepsilon}{16} = \frac{10001}{2937888} > 0. \tag{27}$$

Proof. Every source monomial of A_β has positive degree. Since $0 < \rho \leq 1$, $|A_\beta(s)| \leq \sigma_\beta\rho$, proving $U_\beta \geq 0$. Also $|t^\beta| \leq \rho^{|\beta|}$, proving $V_\beta \geq 0$.

Let g and r be the coefficient norms of G and T, and write $M = \sum_\beta \sigma_\beta$. The nonconstant norm of this frame is exactly $g + r + M \leq B_*$. The losses in p obey

$$p(s) \geq \gamma_0/2 - \rho g - \tfrac{3}{2}B.$$

The corresponding bound for q is $q(t) \geq \gamma_0/2 - \rho r - \frac{3}{2}B$. Because $|\beta| \geq 1$ and $\rho \leq 1$, we have $B \leq \rho M$. Thus either lower bound is at least

$$\gamma_0/2 - \tfrac{3}{2}\rho(g + r + M) > \varepsilon/2 - 3\varepsilon/16 = 5\varepsilon/16.$$

The stated weak inequality follows. This proof uses finite polynomial inequalities, with no numerical approximation or compactness step. □

2.6 Explicit blocks and tensor certificates

For a given frame ℓ_k, regard $\ell_k(h)$ and $\ell_k(t)$ as the same rational linear form evaluated in its respective variable group. There are two pure separated pairs and one pair for each mixed group:

$$(U, V) = (p_k, 1), \quad (1, q_k), \quad (U_{k,\beta}, V_{k,\beta}). \tag{28}$$

For each such pair define

$$\psi(t) = \sqrt{V(t)} \begin{pmatrix} 1 \\ \ell_k(t) \end{pmatrix}, \qquad Q(s) = U(s) \begin{pmatrix} \ell_k(h)^2 & -\ell_k(h) \\ -\ell_k(h) & 1 \end{pmatrix}. \tag{29}$$

Lemma 2.2 makes ψ well defined and Q positive semidefinite. Moreover,

$$\psi(t)^T Q(s) \psi(t) = U(s)V(t)\big(\ell_k(t) - \ell_k(h)\big)^2. \tag{30}$$

Combining (26), (19), and (12), and then imposing all six contact equations, proves Theorem 2.1. No truncated Taylor term or contact remainder survives this assembly.

The evaluated grouping gives 735 mixed pairs. The two pure pairs for each of 25 frames add 50 more pairs. Therefore

$$735 + 50 = 785 \text{ blocks}, \qquad 2 \cdot 785 = 1570 \text{ listed features}. \tag{31}$$

Repeated features are retained with their block labels; no unsupported claim of feature linear independence is made.

The same construction works over independently labelled real closed field extensions $K_c, K_t \supset \mathbb{R}$. Put (c, θ) in the declared source domain over K_c, and t in the testing box over K_t. Every separated term is the square in $K_c \otimes_{\mathbb{R}} K_t$ of

$$(\sqrt{U(s)} \otimes \sqrt{V(t)})(1 \otimes \ell_k(t) - \ell_k(h) \otimes 1). \tag{32}$$

Square roots are taken separately in their own real closed fields. This is a local tensor sum-of-squares statement on the displayed domains. It does not extend these domains or provide the missing global finite cover.

2.7 Complete product compiler and seven source columns

For every block in (29), its three symmetric feature products are exactly

$$\psi_0^2 = V, \qquad \psi_0\psi_1 = V\ell_k, \qquad \psi_1^2 = V\ell_k^2. \tag{33}$$

Their supports have 237 distinct monomials. Fix the lexicographic ordering of their five-component exponent vectors and write these monomials as $m_0(t), \ldots, m_{236}(t)$. The maximum degree is six. Define the complete coefficient array $p_{b,ij,k}$ by

$$\psi_{b,i}(t)\psi_{b,j}(t) = \sum_{k=0}^{236} p_{b,ij,k} m_k(t), \qquad 0 \le i \le j \le 1. \tag{34}$$

Every nonzero coefficient is supplied; there are 27,747 records. The number 237 is an ambient monomial count, not a separately asserted rank of the product span.

For the target retain all seven source variables in (2); set only $Z = 1$ and $z = \theta_* + t$. Because the parent is linear in these seven variables, there is a complete rational array $T_{k\nu}$ such that

$$\widetilde{P}_{(d,n,m,C,A,D,\alpha)}(\theta_* + t, 1) = \sum_{k=0}^{236} \sum_{\nu=0}^{6} T_{k\nu}\lambda_\nu m_k(t), \quad \lambda = (d, n, m, C, A, D, \alpha). \tag{35}$$

All seven columns lie in the listed ambient monomial space. Their 295 nonzero coefficients are delivered, including the full d column.

The resulting conic coefficient matching system is

$$\sum_{b=0}^{784} \sum_{0 \le i \le j \le 1} (2 - \delta_{ij}) p_{b,ij,k} Q_{b,ij} = \sum_{\nu=0}^{6} T_{k\nu}\lambda_\nu, \quad 0 \le k \le 236, \qquad Q_b \succeq 0. \tag{36}$$

There are 2,355 scalar upper-triangular Gram variables and 237 coefficient equations. The factor two on the off-diagonal entries is part of the trace pairing and must be retained. For a coefficient functional η on the ambient monomials, the corresponding dual block has entries

$$M_b(\eta)_{ij} = \sum_{k=0}^{236} p_{b,ij,k}\eta_k. \tag{37}$$

Equations (36) and (37) are evaluated local coefficient data. They are not an evaluated projection onto the eight original quantum coordinates. Nor is it asserted that every polynomial nonnegative on the box belongs to this particular finite certificate cone.

2.8 Files, exact replay, and verification limits

The reconstruction has no external Python dependency. Use Python 3.10 or later and run the following commands in this file's directory:

```
python regular_c2_build.py
python regular_c2_check.py
```

The builder uses exact rational sparse polynomial arithmetic and writes deterministically ordered JSON. The checker uses a separately written dictionary implementation and does not import the builder. The three coefficient files are:

- `regular_c2_frames.json`: all six contact equations, the source Jacobian, both Taylor-matrix polynomial representations, the exact LDL data, all 25 frame coefficient polynomials, every shifted positive factor and remainder, and the complete block indexing.
- `regular_c2_feature_list.json`: all 1,570 feature entries with rational polynomial radicands and multipliers, plus all 785 source Gram factorizations Uvv^T.
- `regular_c2_product_map.json`: all 27,747 symmetric product records, all 295 target records, all seven full source-column polynomials, and the shared monomial indexing.

The fresh builder receipt is `regular_c2_build_verification.json`; the independently implemented replay receipt is `regular_c2_independent_verification.json`. Each records the actual SHA-256 hashes of the data read or generated. No status from a lost archive is used as evidence of a successful replay.

The independent checker reconstructs the complete parent directly, derives the contact equations and their nonzero minor, checks regularity over the contact box, checks the half-Hessian and the LDL orientation, reconstructs the prescribed polynomial Taylor allocation, and verifies all 25 frame and separation identities. It checks the 50 strictly positive remainder bounds and each mixed-factor bound. Its final parent-minus-contact Taylor remainder has exactly 1,193 nonzero monomials and agrees with the assembled frame certificate as a complete coefficient dictionary. It then reconstructs every feature, source Gram factor, product coefficient, and seven-source target coefficient. Encoding checks are reported separately from mathematical identity groups; a large assertion count is not a count of independently proved theorems.

This is a reproducible exact algebra check together with the analytic sign and assembly proof above. The independent implementation was produced within the same model-assisted reconstruction process. It is not external peer review and is not proof-assistant formal verification.

2.9 Boundary of the result

This record completes the stated regular-C2 local coefficient extraction. It retains the exact contact equations and all state and angular terms of the complete normalized parent. Its testing radicals are certified on the stated box. To obtain a global lift extraction one must still assemble an appropriate finite global source and testing cover, handle every boundary and projective chart with compatible data, and evaluate the final pencil and projection in the original eight coordinates. Those steps are not consequences of the local coefficient counts. No new completion of the global E1–E3 checklist is claimed here.

3 Separated coefficients and a denominator-free local construction

The constructions in this section explain how an already known local quadratic identity can be converted into finitely many fixed testing features. They do not supply the missing global atlas of the quantum set. The distinction is a quantifier issue: a construction on one rectangle must still be extended to, or combined with a finite cover of, every source and every testing point.

3.1 An explicit separation identity for a positive coefficient

Suppose

$$\gamma(x,y)=\sum_{i=1}^{n}a_i(x)b_i(y),\qquad \alpha_i=a_i(x_0),\quad \beta_i=b_i(y_0),\qquad c=\sum_i\alpha_i\beta_i>0.$$

Write $a_i=\alpha_i+A_i$ and $b_i=\beta_i+B_i$. Choose a positive rational number δ such that

$$3n\delta^2+2\delta\max\left\{\sum_i|\alpha_i|,\sum_i|\beta_i|\right\}<c. \tag{38}$$

Such a choice exists because the left side tends to zero with δ. Work on a product neighborhood where $|A_i(x)|\le\delta$ and $|B_i(y)|\le\delta$ for every i. Define

$$\begin{aligned}p(x)&=\frac{c-n\delta^2}{2}+\sum_i(\beta_i-\delta)A_i(x),\\ q(y)&=\frac{c-n\delta^2}{2}+\sum_i(\alpha_i-\delta)B_i(y).\end{aligned} \tag{39}$$

Then the following identity is exact:

$$\boxed{\gamma(x,y)=p(x)+q(y)+\sum_i\big(A_i(x)+\delta\big)\big(B_i(y)+\delta\big).} \tag{40}$$

Indeed, the $n\delta^2$ terms cancel those in $p+q$, the terms linear in A_i and B_i reduce to β_iA_i and α_iB_i, and the remaining terms are A_iB_i. Moreover,

$$\begin{aligned}p(x)&\ge\frac{c}{2}-\frac{3n\delta^2}{2}-\delta\sum_i|\beta_i|>0,\\ q(y)&\ge\frac{c}{2}-\frac{3n\delta^2}{2}-\delta\sum_i|\alpha_i|>0.\end{aligned}$$

The other factors in (40) are nonnegative. Thus a positive coefficient in an algebraic tensor product is, locally, an explicit sum of $n+2$ products of nonnegative one-factor functions. No inverse involving both factors is introduced.

For example, on $|x|,|y|\le1/4$,

$$1+xy=(x+1/4)(y+1/4)+(15/32-x/4)+(15/32-y/4). \tag{41}$$

The last two factors are at least $13/32$.

Conversion into fixed blocks of order two. For any separated nonnegative coefficient $a(x)b(y)$ and any fixed scalar function ℓ, put

$$v(y)=\sqrt{b(y)}\begin{pmatrix}1\\ \ell(y)\end{pmatrix},\qquad Q(x)=a(x)\begin{pmatrix}\ell(x)^2 & -\ell(x)\\ -\ell(x) & 1\end{pmatrix}.$$

Then $Q(x)\succeq0$ and

$$v(y)^TQ(x)v(y)=a(x)b(y)\big(\ell(y)-\ell(x)\big)^2. \tag{42}$$

The two entries of v are fixed functions of the testing variable. Applying this construction to the $n+2$ terms in (40) gives $n+2$ blocks of order two for $\gamma(x,y)(\ell(y)-\ell(x))^2$. If the factors are evaluated in independently labelled real closed fields, their square roots are taken in their respective fields. Equation (42) is therefore also an identity in the corresponding tensor product. It uses no identification of the two field copies.

3.2 The smooth-diagonal statement and its hypotheses

Let $Y \subset \mathbb{R}^m$ be a smooth affine real algebraic set near a point p, with coordinate ring B and ambient coordinate column z. Write

$$\Delta z = z \otimes 1 - 1 \otimes z, \qquad J = \ker(B \otimes_{\mathbb{R}} B \longrightarrow B).$$

The homomorphism is restriction to the diagonal. Suppose an exact identity has already been supplied:

$$\tau = \Delta z^T H \Delta z, \qquad H = H^T \in \mathrm{Mat}_m(B \otimes B), \tag{43}$$

and suppose the restriction of $H(p,p)$ to T_pY is positive definite. Membership $\tau \in J^2$ gives a representation of the form (43), because J is generated by the coordinate differences; the positive tangential residue is a separate hypothesis. It is not implied by membership in J^2.

Under these hypotheses, there is a product neighborhood of (p,p) on which τ has a finite fixed-feature representation using only blocks of order two. This representation can be constructed without inverting a mixed function of the two points.

Step 1: remove only the normal degeneracy. Choose defining equations $g_1, \dots, g_r$ whose differentials span the conormal space at p. Telescopic polynomial divided differences give a matrix $A \in \mathrm{Mat}_{r \times m}(B \otimes B)$ with

$$A\Delta z = 0, \qquad A(p,p) = Dg(p), \qquad \ker A(p,p) = T_pY.$$

For example, every polynomial difference $g(z) - g(z')$ is a sum of the coordinate differences multiplied by polynomial divided differences. In $B \otimes B$ that polynomial difference is zero.

There is a positive integer t for which

$$H_* = H(p,p) + tA(p,p)^T A(p,p)$$

is positive definite on $\mathbb{R}^m$. To see this directly, decompose $\mathbb{R}^m = T_pY \oplus (T_pY)^\perp$. The upper left block of $H(p,p)$ is positive definite. On the normal summand, $A(p,p)^T A(p,p)$ is positive definite. The Schur complement of the upper left block becomes positive for all sufficiently large t. Adding tA^TA does not change (43), since $\Delta z^T A^T A \Delta z = 0$ exactly.

Step 2: use a constant change of coordinates. Choose a real invertible constant matrix C with $H_* = C^T C$ and put

$$f = Cz, \qquad \widehat{H} = C^{-T}(H + tA^TA)C^{-1}.$$

Then $\widehat{H}(p,p) = I_m$ and $\tau = \Delta f^T \widehat{H} \Delta f$. The inverses here are inverses of a fixed real matrix, not inverses of functions on $Y \times Y$.

Step 3: use a fixed finite frame. Set $\varepsilon = 1/(4m)$. For every symmetric matrix K and vector $d \in \mathbb{R}^m$, the polynomial identity

$$\begin{aligned} d^T K d = &\sum_i (K_{ii} - 2(m-1)\varepsilon) d_i^2 \\ &+ \sum_{i<j} \left(\varepsilon + \frac{K_{ij}}{2}\right)(d_i + d_j)^2 \\ &+ \sum_{i<j} \left(\varepsilon - \frac{K_{ij}}{2}\right)(d_i - d_j)^2 \end{aligned} \tag{44}$$

follows by expansion. Apply it with $K = \widehat{H}$ and $d = \Delta f$. At (p,p) its coefficients are $(m+1)/(2m)$ on the m coordinate squares and $1/(4m)$ on the pair squares. All are strictly positive. They remain positive on a sufficiently small product neighborhood.

There are m^2 fixed frame functions: f_i and $f_i + f_j, f_i - f_j$ for $i < j$. Each coefficient lies in $B \otimes B$ and therefore has a finite separated expansion. Apply (40) to each coefficient, shrink the two neighborhoods finitely many times, and use (42). This proves the claimed local construction.

What denominators are permitted. Pure-factor inverses can be handled by adjoining, to that factor, a coordinate satisfying an equation such as $rh = 1$, on the locus where $h \neq 0$. This is still a one-factor construction. The analogous move for a mixed denominator need not have finite separation rank. For instance,

$$K(x, y) = \frac{1}{1 - xy}$$

has infinite separation rank on any product of open intervals about zero. If it were a sum of N separated functions, choose $N+1$ distinct nonzero x_j in the first interval. The functions $1/(1-x_j y)$ would lie in an N-dimensional space. They are linearly independent: a linear relation, multiplied by $\prod_j (1 - x_j y)$, is a polynomial identity, and evaluation at $y = 1/x_j$ kills all but the jth coefficient. This contradiction does not require those poles to lie in the original interval.

The theorem above therefore requires the actual J^2 identity in the indicated tensor product, not a representation obtained by an uncontrolled mixed rational inversion. It also leaves the global finite-cover question open.

4 Exact extraction of the displayed B_0 coefficient module

The recovered August 30 source displays, in its equation (12.1), a matrix with four scalar coefficients. The August 31 correction R5 specifies its factor typing: $A, T, p, w, h, \varepsilon_0$ belong to the point factor, and U, V belong to the support factor. We retain that typing. For the algebraic calculation below, c_i simply denote the four coefficient-factor scalars; the phrase "testing variables" means (U, V, z) in this module, without asserting a new physical source-to-point identification.

Let $e_0, \dots, e_3$ be the standard basis of $\mathbb{R}^4$ and $\omega = (U, -1, 0, -UV)^T$. Define

$$\begin{aligned} S_c(U, V) = {} & c_1(1 + U^2) e_0 e_0^T + c_2(1 + V^2)\omega\omega^T \\ & + c_3(1 + U^2)(1 + V^2) e_1 e_1^T + c_4 U^2 e_2 e_2^T. \end{aligned} \tag{45}$$

Then

$$S_c(U, V) \succeq 0 \text{ for every real } U, V \quad \Longleftrightarrow \quad c_1, c_2, c_3, c_4 \geq 0. \tag{46}$$

The reverse implication follows from the displayed sum. The forward implication follows from four rational tests:

(U, V)	state z	$z^T S_c(U, V) z$
$(0, 0)$	e_0	c_1
$(1, 1)$	e_3	$2c_2$
$(1, 1)$	$(0, 1, 0, -1)^T$	$4c_3$
$(1, 0)$	e_2	c_4

Thus this four-dimensional coefficient cone has the evaluated pencil $\operatorname{diag}(c_1, c_2, c_3, c_4) \succeq 0$, with order four and no auxiliary variables.

For $z = (z_0, z_1, z_2, z_3)^T$, a fixed nine-function certificate is

$$\begin{aligned} v = \big(z_0,\ & Uz_0,\ Uz_0 - z_1 - UVz_3,\ UVz_0 - Vz_1 - UV^2 z_3, \\ & z_1,\ Uz_1,\ Vz_1,\ UVz_1,\ Uz_2\big)^T, \end{aligned}$$

and the exact identity is

$$z^T S_c z = v^T \operatorname{diag}(c_1, c_1, c_2, c_2, c_3, c_3, c_3, c_3, c_4) v.$$

For the original displayed coefficients,

$$\begin{aligned} p &= 1 - A - LT^2, & w &= 1 - LT^2, & h &= 1 - L(1 + T^2), \\ c_1 &= \frac{AT(1 + T^2)p}{w}, & c_2 &= A\varepsilon_0, & c_3 &= \varepsilon_0 p, \\ c_4 &= \frac{A\varepsilon_0(1 + T^2)wp}{h}. \end{aligned}$$

At the rational strict-chamber point $(T, L, A, \varepsilon_0) = (1, 2/5, 1/5, 1)$, these become

$$(p, w, h) = (2/5, 3/5, 1/5), \qquad (c_1, c_2, c_3, c_4) = (4/15, 1/5, 2/5, 12/25).$$

The calculation also gives $L(T^2 + 2) - 1 = 1/5 > 0$.

A precise conditional congruence envelope. Every displayed feature belongs to

$$\operatorname{span}\{1, U, V, UV, UV^2\} \otimes \operatorname{span}\{z_0, z_1, z_2, z_3\}.$$

Consequently, after a 4×4 state congruence $z = R\widetilde{z}$ whose entries depend only on the coefficient factor, the twenty fixed functions

$$\{\widetilde{z}_i, U\widetilde{z}_i, V\widetilde{z}_i, UV\widetilde{z}_i, UV^2\widetilde{z}_i : 0 \le i < 4\}$$

contain every transformed feature. A congruence depending arbitrarily on U, V is not covered by this conclusion. This containment also does not assert that an order-two block bound survives an arbitrary change of state coordinates.

The recovered text describes a pure-point radial congruence and a common positive clearing before (12.1), but does not print those maps. Therefore the exact module, its orthant, and the conditional twenty-function envelope are complete; the restoration of this module to the full original-coordinate slack remains an explicit missing input.

5 Reconstructed event-contact cofactors

This calculation starts from the canonical matrix in equation (4.2) of the recovered August 30 source. Restrict to the kernel of its λ row by writing

$$\psi = (-vy_3, y_3, y_1, y_2)^T,$$

and set the two extra coefficients $\alpha = \beta = 0$. The restricted quadratic form has coefficients

$$\begin{aligned} G_d &= 2\big[(D_v + u^2v^2)y_1^2 - 2u^2vy_1y_2 + u^2y_2^2 + D_vy_3^2 + 2uD_vy_2y_3\big], \\ G_1 &= 4D_uD_vy_3^2, \qquad G_2 = 4(uvy_1 - uy_2 - D_vy_3)^2, \\ G_3 &= 4D_v(y_1 + uvy_3)^2, \qquad G_4 = 4D_uD_vy_1^2, \qquad G_5 = 4(y_1 + vy_2)^2, \end{aligned} \tag{47}$$

where $D_u = 1 + u^2$ and $D_v = 1 + v^2$. These six restrictions have been checked directly against the canonical row vectors. Substituting $u = u_1/u_0$, $v = v_1/v_0$ and multiplying the five positive G_i by $u_0^2v_0^4$ gives exactly four times the five homogeneous forms in the recovered equation (11.0). This is an algebraic homogenization identity, including its polynomial extension to projective endpoints.

On the state chart $y_2 = 1$, put $a = y_1$ and $b = y_3$. Let $G = (G_d, G_1, \dots, G_5)$ after this substitution and form

$$J = \begin{pmatrix} G \\ \partial_u G \\ \partial_v G \\ \partial_a G \\ \partial_b G \end{pmatrix}, \qquad X_j = (-1)^j \det J_{\widehat{j}}, \quad 0 \le j \le 5.$$

Here $J_{\widehat{j}}$ is the five-column matrix obtained by deleting column j, and indexing starts at zero. Laplace expansion gives $JX = 0$ identically. Direct exact calculation over $\mathbb{Q}[u, v, a, b]$ gives the common factor

$$g = 8192\, b(1 + u^2)(1 + v^2)^3.$$

After division by g, the primitive polynomial vector $\widetilde{X} = X/g$ has component degrees

$$(15, 13, 13, 11, 15, 17)$$

and respectively

$$(59, 40, 43, 36, 42, 38)$$

nonzero monomials. Its component gcd is one and $J\widetilde{X} = 0$ is checked as five exact polynomial identities. All coefficients, the six defining forms, and the thirty Jacobian entries are printed as machine-readable rational polynomial records in `diagonal/EVENT_CONTACT_EXACT.json`.

These cofactors parametrize stationary solutions of the restricted contact equations wherever the contact matrix has rank five. They do not by themselves prove componentwise nonnegativity, physical support validity, a full four-state kernel equation, or completeness at rank drop and projective boundaries. Those additional claims must be proved separately before this vector can be used as a global source parametrization.

6 The finite-cover assembly that remains to be instantiated

The final passage from local fixed features to an affine pencil is elementary once the necessary global data have actually been supplied. For clarity, the statement below lists those data rather than presupposing them.

Let T be a testing set with a map $\Phi : T \to \mathbb{R}^d$, and let $C = \operatorname{conv} \Phi(T)$ be closed and full dimensional. Let S index every affine functional nonnegative on C, including its constant. Suppose that finitely many rectangles $U_i \times V_i \subset S \times T$ cover all of $S \times T$. On each rectangle, assume a supplied exact slack identity

$$f_s(\Phi(t)) = \sum_k v_{ik}(t)^T Q_{ik}(s) v_{ik}(t), \qquad Q_{ik}(s) \succeq 0 \quad (s \in U_i), \tag{48}$$

with finitely many fixed vector functions v_{ik} of order at most two. All normalizations, coordinate changes, and clearing factors must already have been restored in this identity.

Step 1: convert the rectangle cover into fixed functions. Partition S by its membership patterns in the finitely many U_i, and partition T by its membership patterns in the V_i. If A and B are nonempty cells of these partitions, the covering property provides an $i = i(A, B)$ such that $A \times B \subset U_i \times V_i$. This follows because membership in every U_i, V_i is constant on a cell. Choose one such index once for each pair of cells.

For each pair (A, B), extend $v_{i(A,B),k}$ by zero outside B. Extend its source matrix by zero outside A. For a fixed source there is exactly one source cell, while every test lies in exactly one test cell. Summing the resulting finitely many identities gives a global representation with fixed testing functions and positive source matrices. The indicator functions merely select cells; they do not enlarge the vector order. If the original data and cells are semialgebraic, these zero extensions are semialgebraic as well.

Step 2: form the finite product space. Let v_j denote the resulting finite family of vector functions. Define the real vector space

$$W = \operatorname{span}_{\mathbb{R}}\{1, \Phi_1, \dots, \Phi_d, (v_j)_a (v_j)_b : j, a, b\}.$$

Choose an actual basis $w_0 = 1, w_1 = \Phi_1, \dots, w_d = \Phi_d, w_{d+1}, \dots, w_D$. Full dimensionality ensures independence of the first $d + 1$ functions. Record every product expansion

$$(v_j)_a (v_j)_b = \sum_{k=0}^{D} P_{j,ab,k} w_k.$$

Only this finite linear span is needed; multiplication closure of W is not a requirement.

Step 3: write and prove the pencil. For $x \in \mathbb{R}^d$ and $y \in \mathbb{R}^{D-d}$, put

$$M_j(x,y)_{ab} = P_{j,ab,0} + \sum_{\nu=1}^{d} x_\nu P_{j,ab,\nu} + \sum_{k=d+1}^{D} y_k P_{j,ab,k}. \tag{49}$$

Then

$$C = \{x : \exists y,\ M_j(x,y) \succeq 0 \text{ for every } j\}.$$

For the forward inclusion, if x is a finite convex combination of $\Phi(t)$, let L be the same convex combination of evaluations on W. Each $M_j = L(v_j v_j^T)$ is positive semidefinite, $L(1) = 1$, and $L(\Phi_\nu) = x_\nu$.

For the reverse inclusion, a feasible collection (x, y) defines the linear functional L on the chosen basis of W. If $x \notin C$, closed convex separation gives an affine f_s nonnegative on C with $f_s(x) < 0$. Apply L to its global identity assembled in Step 1. The left side is $f_s(x)$, while the right side is a sum of $\operatorname{tr}(Q_j(s)M_j(x,y)) \geq 0$, a contradiction. Both inclusions are thus proved once the hypotheses and the actual coefficient tables have been evaluated.

For the present full eight-coordinate quantum problem, those global inputs have not all been supplied. The local constructions in this release give usable pieces of the required data. They do not replace the missing restoration maps, exceptional and boundary coverage, or the finite all-source/all-test cover. Consequently the global feature list, global evaluated pencil, and verification of both global inclusions remain open.

Fresh verification records. The script `diagonal/regenerate_diagonal.py` regenerates the exact coefficient data and four algebraic verification groups with SymPy 1.14.0. The corresponding receipt is `diagonal/DIAGONAL_VERIFICATION.json`. Its checks corroborate the displayed identities and restricted cofactor computations. They are not a proof-assistant verification of the analytical arguments and do not certify the missing global hypotheses.

7 Exact tilted–CHSH projection and support family

7.1 Coordinates, scope, and the extracted set

Write a behavior in the order

$$q = (a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}) \in \mathbb{R}^8.$$

Here $a_x = \langle A_x \rangle$, $b_y = \langle B_y \rangle$, and $E_{xy} = \langle A_x B_y \rangle$. The parties' observables commute, and after a finite-dimensional projective dilation they satisfy $A_x^2 = B_y^2 = \mathbb{I}$. Define

$$t = a_0, \qquad s = E_{00} + E_{01} + E_{10} - E_{11}.$$

The full quantum set $\mathcal{Q}_{222}$ is the collection of such behaviors obtained from finite-dimensional tensor-product quantum models, including mixed states and binary positive operator valued measurements.

Theorem 7.1 (Exact tilted projection)**.** *The image of $\mathcal{Q}_{222}$ under $q \mapsto (t, s)$ is*

$$\mathcal{T} = \{(t,s) \in \mathbb{R}^2 : -1 \leq t \leq 1, \quad t^2 + s^2/4 \leq 2\}.$$

Its support function in the direction (α, ϵ), where $\epsilon \in \{-1, 1\}$, is

$$h(\alpha) = \max_{(t,s)\in\mathcal{T}} (\alpha t + \epsilon s) = \begin{cases} \sqrt{8 + 2\alpha^2}, & |\alpha| \leq 2, \\ 2 + |\alpha|, & |\alpha| \geq 2. \end{cases} \tag{50}$$

The two formulas agree at $\alpha = \pm 2$.

The theorem is proved directly below. Its purpose is to identify precisely the domain for which the following finite extraction is complete. A lift written in the original eight coordinates describes the inverse image of $\mathcal{T}$ under this projection, not $\mathcal{Q}_{222}$ itself.

7.2 The quantum upper bound and attainment

7.2.1 Reflection identity

Let

$$\begin{aligned} S &= A_0(B_0 + B_1) + A_1(B_0 - B_1), \\ S' &= A_0(B_0 - B_1) + A_1(B_0 + B_1), \\ T_\alpha &= \alpha A_0 + S, \qquad R_\alpha = \alpha A_1 - S'. \end{aligned}$$

Using only the reflection relations and cross-party commutation gives

$$T_\alpha^2 + R_\alpha^2 = (8 + 2\alpha^2)\mathbb{I}. \tag{51}$$

For completeness, the terms independent of α satisfy $S^2 + (S')^2 = 8\mathbb{I}$. The terms linear in α cancel because

$$\begin{aligned} \{A_0, S\} &= 2(B_0 + B_1) + \{A_0, A_1\}(B_0 - B_1), \\ \{A_1, S'\} &= 2(B_0 + B_1) + \{A_0, A_1\}(B_0 - B_1). \end{aligned}$$

The quadratic terms total $2\alpha^2\mathbb{I}$. With $\beta = \sqrt{8 + 2\alpha^2} > 0$, equation (51) implies

$$\beta\mathbb{I} - T_\alpha = \frac{(\beta\mathbb{I} - T_\alpha)^2 + R_\alpha^2}{2\beta} \succeq 0. \tag{52}$$

This is the first tilted–CHSH sum of squares of Bamps and Pironio, equation (27) in the arXiv version [1]. The regenerated checker expands both sides in reduced noncommutative words and checks equality exactly.

Every binary observable before dilation is a self-adjoint contraction C. The block operator

$$\widehat{C} = \begin{pmatrix} C & \sqrt{\mathbb{I} - C^2} \\ \sqrt{\mathbb{I} - C^2} & -C \end{pmatrix}$$

is a reflection on a doubled local space and compresses to C under the same fixed embedding. Applying this separately to each observable on each party preserves all of the eight coordinates. Thus the bound covers arbitrary binary measurements. A mixed state may be used directly by taking traces of the positive operator in (52).

We obtain $\alpha t + s \le \sqrt{8 + 2\alpha^2}$ for every real α, as well as $|t| \le 1$. For a given $t \in [-1, 1]$, take

$$h_t = \sqrt{2 - t^2}, \qquad \alpha = \frac{2t}{h_t}.$$

Then $|\alpha| \le 2$ and the preceding bound gives

$$s \le \frac{4}{h_t} - \frac{2t^2}{h_t} = 2h_t.$$

Flipping both Bob outcomes gives $-s \le 2h_t$. Consequently $(t, s) \in \mathcal{T}$.

7.2.2 Explicit attainment and both boundary points

Fix $t \in [-1, 1]$ and set $r = \sqrt{1 - t^2}$ and $h_t = \sqrt{2 - t^2}$. On two qubits choose

$$\begin{aligned} |\psi_t\rangle &= \sqrt{\frac{1+t}{2}}\,|00\rangle + \sqrt{\frac{1-t}{2}}\,|11\rangle, \\ A_0 &= Z, \qquad A_1 = X, \\ B_0 &= \frac{Z + rX}{h_t}, \qquad B_1 = \frac{Z - rX}{h_t}. \end{aligned}$$

The anticommutation of X and Z implies $B_0^2 = B_1^2 = \mathbb{I}$. Direct evaluation gives

$$q_t = \left(t, 0, \frac{t}{h_t}, \frac{t}{h_t}, \frac{1}{h_t}, \frac{1}{h_t}, \frac{1-t^2}{h_t}, -\frac{1-t^2}{h_t}\right). \tag{53}$$

Its projection is $(t, 2h_t)$. Flipping both Bob observables gives $(t, -2h_t)$ without changing t. A classical mixture of those two finite-dimensional models realizes every intermediate value $s \in [-2h_t, 2h_t]$. The construction retains $t = \pm 1$: the state is then a computational-basis product state, $r = 0$, and $h_t = 1$. No division by a vanishing quantity occurs. This proves both inclusions in Theorem 7.1.

To compute the support function, maximize

$$\alpha t + 2\sqrt{2 - t^2} \qquad (-1 \le t \le 1).$$

Its unconstrained stationary point is

$$t_* = \frac{\sqrt{2}\,\alpha}{\sqrt{4 + \alpha^2}}.$$

This point lies in the interval exactly when $|\alpha| \le 2$, and its value is $\sqrt{8 + 2\alpha^2}$. For $|\alpha| \ge 2$, the maximum occurs at $t = \operatorname{sgn}(\alpha)$ and is $2 + |\alpha|$. The symmetry $s \mapsto -s$ covers both signs of ϵ.

7.3 One fixed collection of testing functions

Let $K_t \supseteq \mathbb{R}$ be a real closed extension and $(t, s) \in \mathcal{T}_{K_t}$. All square roots below mean the nonnegative root in K_t. Define

$$\begin{aligned} u_1 &= \sqrt{\sqrt{2} + t}, & u_2 &= \frac{s}{2u_1}, & u_3 &= \sqrt{\sqrt{2} - t - u_2^2}, \\ u_4 &= \sqrt{1 - t}, & u_5 &= \sqrt{1 + t}. \end{aligned} \tag{54}$$

These functions are defined throughout $\mathcal{T}_{K_t}$. In fact,

$$\sqrt{2} + t \ge \sqrt{2} - 1 > 0,$$

and

$$(\sqrt{2} + t)(\sqrt{2} - t - u_2^2) = 2 - t^2 - s^2/4 \ge 0.$$

Their definition depends only on the testing point, not on a supporting functional. The four fixed square spaces are

$$U_1 = \operatorname{span}_{\mathbb{R}}(u_1, u_2), \quad U_2 = \mathbb{R}u_3, \quad U_3 = \mathbb{R}u_4, \quad U_4 = \mathbb{R}u_5.$$

The constant function is already available as a sum of squares:

$$1 = \frac{u_4^2 + u_5^2}{2}. \tag{55}$$

7.3.1 Separated source and testing fields

Let $K_c \supseteq \mathbb{R}$ be another real closed extension. For $\alpha \in K_c$ and $\epsilon \in \{-1, 1\}$ define in the source field

$$\begin{aligned} \gamma &= \sqrt{4 + \alpha^2}, & \beta &= \sqrt{2}\,\gamma, \\ v_0 &= \sqrt{\frac{\gamma - \alpha}{2}}, & v_1 &= -\epsilon\sqrt{\frac{\gamma + \alpha}{2}}. \end{aligned}$$

Here $\gamma > |\alpha|$, and

$$v_0^2 + v_1^2 = \gamma, \quad v_0^2 - v_1^2 = -\alpha, \quad v_0 v_1 = -\epsilon.$$

The last identity uses $\gamma^2-\alpha^2=4$ and the specified root signs. Expanding squares in the commutative tensor-product ring gives

$$\begin{aligned}&\beta\otimes 1-\alpha\otimes t-1\otimes \epsilon s\\&\quad=(v_0\otimes u_1+v_1\otimes u_2)^2+(v_1\otimes u_3)^2\quad\text{in } K_c\otimes_{\mathbb{R}} K_t.\end{aligned}\tag{56}$$

Indeed, the right side equals

$$v_0^2\otimes(\sqrt{2}+t)+2v_0v_1\otimes(s/2)+v_1^2\otimes(\sqrt{2}-t).$$

The displayed source identities give the left side. No product of a source variable with a testing variable is identified with a product inside either field. There is no disjointness assumption: K_c and K_t may be two separately labelled copies of the same field.

For $|\alpha|\le 2$, equation (56) certifies the optimal support slack. For $|\alpha|\ge 2$, set $\sigma=\operatorname{sgn}(\alpha)$. The exact endpoint completion is

$$\begin{aligned}2+|\alpha|-\alpha t-\epsilon s&=4-2\sigma t-\epsilon s\\&\quad+(|\alpha|-2)(1-\sigma t).\end{aligned}\tag{57}$$

The first term has the certificate (56) at $\alpha=2\sigma$. The second is a nonnegative source multiple of u_4^2 if $\sigma=1$, or u_5^2 if $\sigma=-1$. Its coefficient square root is taken in K_c. Equality $|\alpha|=2$ merely makes that coefficient zero. The case $\alpha=0$ is already contained in (56). Thus every real tilt, including every endpoint and unbounded source tilts in real closed extensions, uses these same fixed testing spaces.

7.3.2 All affine nonnegative functions on this projection

Let $f=c_0+c_1t+c_2s$ be nonnegative on $\mathcal{T}$ (or its transfer), with coefficients in the source field. If $c_2\neq 0$, put

$$\lambda=|c_2|,\quad \alpha=-c_1/\lambda,\quad \epsilon=-\operatorname{sgn}(c_2).$$

Equation (50) implies $c_0-\lambda h(\alpha)\ge 0$, and

$$f=(c_0-\lambda h(\alpha))+\lambda\big(h(\alpha)-\alpha t-\epsilon s\big).$$

The first term is handled by (55); the second uses (56) and, where necessary, (57). If $c_2=0$, nonnegativity is equivalent to $c_0\ge|c_1|$, and

$$f=(c_0-|c_1|)+|c_1|(1+\operatorname{sgn}(c_1)t)$$

when $c_1\neq 0$; the constant case is immediate. This verifies coverage of all affine supports of $\mathcal{T}$, rather than only a bounded subinterval of the tilt parameter. It does not assert coverage of all affine supports of $\mathcal{Q}_{222}$.

7.4 Product table and every affine pencil coefficient

Set $y=u_2^2$. The products within the four square spaces have the following complete table in the linear coordinates $(1,t,s,y)$.

Product	Value
u_1^2	$\sqrt{2}+t$
u_1u_2	$s/2$
u_2^2	y
u_3^2	$\sqrt{2}-t-y$
u_4^2	$1-t$
u_5^2	$1+t$

The evaluated block diagonal pencil is

$$L(t,s,y)=\begin{pmatrix}\sqrt{2}+t & s/2 & 0 & 0 & 0\\ s/2 & y & 0 & 0 & 0\\ 0 & 0 & \sqrt{2}-t-y & 0 & 0\\ 0 & 0 & 0 & 1-t & 0\\ 0 & 0 & 0 & 0 & 1+t\end{pmatrix}. \tag{58}$$

Proposition 7.2 (Both lift inclusions)**.** *The identity*

$$\mathcal{T}=\{(t,s):\exists y\in\mathbb{R},\ L(t,s,y)\succeq 0\}$$

holds. Thus the evaluated lift has $N=5$, $m=1$, and block orders $(2,1,1,1)$.

Proof. If $(t,s)\in\mathcal{T}$, choose $y=s^2/[4(\sqrt{2}+t)]$. The upper-left block is the Gram matrix of (u_1,u_2), hence is positive semidefinite; the other diagonal blocks are u_3^2,u_4^2,u_5^2. Conversely, positivity forces $|t|\le 1$, so $d=\sqrt{2}+t>0$. The Schur complement of the upper-left entry gives $y\ge s^2/(4d)$, while the next scalar block gives $y\le\sqrt{2}-t$. Therefore

$$s^2/4\le(\sqrt{2}+t)(\sqrt{2}-t)=2-t^2,$$

which proves $(t,s)\in\mathcal{T}$. □

Eliminating y gives the direct pencil

$$D(t,s)=\operatorname{diag}\left(\begin{pmatrix}\sqrt{2}+t & s/2\\ s/2 & \sqrt{2}-t\end{pmatrix},1-t,1+t\right). \tag{59}$$

It has $N=4,m=0$, with block orders $(2,1,1)$. Under $|t|\le 1$, the diagonal entries of its 2×2 block are strictly positive, and its determinant is $2-t^2-s^2/4$. Hence $D\succeq 0$ is exactly the defining condition for $\mathcal{T}$.

For the original coordinates substitute $t=a_0$ and $s=E_{00}+E_{01}+E_{10}-E_{11}$ into (58). Every coefficient matrix is specified by the following upper-triangular entries, using zero-based matrix indices. Each off-diagonal entry is mirrored once; the tabulated value is the actual entry, not a doubled Gram coefficient.

Coefficient	Index	Value
constant	$(0,0),(2,2)$	$\sqrt{2},\sqrt{2}$
constant	$(3,3),(4,4)$	$1,1$
a_0	$(0,0),(2,2),(3,3),(4,4)$	$1,-1,-1,1$
E_{00},E_{01},E_{10}	$(0,1)$ in each matrix	$1/2$
E_{11}	$(0,1)$	$-1/2$
y	$(1,1),(2,2)$	$1,-1$

All unlisted entries, including the matrices multiplying a_1,b_0,b_1, are zero. There are exactly 14 nonzero upper-triangular records. The eliminated pencil has 12 such records. The two JSON coefficient files encode every number as a pair of rational strings (r_0,r_1) representing $r_0+r_1\sqrt{2}$, where $\sqrt{2}$ is the positive root of X^2-2. No approximate coefficient or numerical reconstruction is used.

7.5 A strict scope witness

Consider the rational behavior

$$q_*=(0,0,0,0,1,1,1/4,-1/4). \tag{60}$$

For outcomes $a, b \in \{-1, 1\}$, its probabilities are

$$p(a, b \mid x, y) = \frac{1 + aa_x + bb_y + abE_{xy}}{4} = \frac{1 + abE_{xy}}{4}.$$

They are nonnegative and normalized for each setting, and the marginal distributions are uniform and nonsignalling. Every CHSH relabeling has an odd number of negative signs among its four correlator coefficients. Enumerating the eight sign patterns gives maximum absolute value $5/2 < 2\sqrt{2}$. Relabeling settings, outcomes, or parties leaves all marginal biases zero. Consequently every standard relabeled single-marginal tilted–CHSH expression has value at most $5/2$, while its quantum support bound satisfies $h(\alpha) \geq 2\sqrt{2}$ for every real α. Thus all those inequalities and all probability inequalities accept q_*.

Nevertheless q_* is nonquantum. In any putative realization, projectively dilate and purify the state to $|\psi\rangle$. The equalities $E_{00} = E_{01} = 1$ give

$$\|(A_0 - B_y)|\psi\rangle\|^2 = 2 - 2E_{0y} = 0 \qquad (y = 0, 1).$$

Here the identity factors on the other local systems are implicit. Subtracting the resulting vector equalities gives $(B_0 - B_1)|\psi\rangle = 0$. It follows that

$$E_{10} - E_{11} = \langle\psi|A_1(B_0 - B_1)|\psi\rangle = 0,$$

contradicting $E_{10} - E_{11} = 1/2$ in (60). This argument applies to mixed states through purification and to general binary measurements through the stated dilation. It is an exact algebraic obstruction, not the result of a numerical infeasibility test.

The witness even lies strictly inside the ellipse at its projected point: $t = 0, s = 5/2$ gives $t^2 + s^2/4 = 25/16 < 2$. Therefore intersecting every standard relabeled tilted-family lift with the probability constraints still does not recover $\mathcal{Q}_{222}$. Global feature extraction, a global affine pencil, and verification of the full-set inclusions remain open in this regeneration.

7.6 Fresh verification and reproducibility

In this directory, run

```
python3 build_and_check_tilted_extraction.py
```

The program uses only Python's standard library. The helper implements rational sparse-polynomial expansion and reduced reflection words; no floating-point arithmetic, optimization solver, or quantifier elimination is used.

The regenerated receipt records 38 named checks. They include the noncommutative reflection and SOS identities, all eight Pauli expectation numerators in (53), the source and testing identities, the endpoint completion, reconstruction of the original-coordinate matrices, both coefficient record counts, the eliminated determinant, exact probability normalization and positivity for the scope witness, and enumeration of all eight CHSH sign patterns. The checker does not mechanize the analytical inequalities, quantum reduction arguments, or proofs over arbitrary real closed extensions. Those arguments are given above. The manifest binds the checker and its generated data by SHA-256.

This newly regenerated component is mathematically scoped and independently replayable. It is not a byte-for-byte restoration of an unavailable historical 549-page release, and its successful replay makes no assertion about the missing global existence proof.

8 A limit of literal Schur-remainder extraction

The complete parent and a scalar expression obtained by eliminating state coordinates have different feature requirements. This section gives an exact example: the literal Schur remainders of one

physical C2 family span an infinite-dimensional vector space, although the complete family already has five fixed testing features. The obstruction therefore concerns one extraction procedure, and supplies no obstruction to a finite lift of the complete parent.

Use the complete quarter-parent P_c in the regular-C2 section, with $D_u = 1 + u^2$, $D_v = 1 + v^2$, state column $\psi = (W, X, Y, Z)^T$, and source order $(d, n, m, C, A, D, \alpha)$. Consider

$$c(t) = \left(1, \frac{7}{8} + t, \frac{9}{8}, \frac{1}{18}, \frac{11}{8}, \frac{9}{8}, \frac{1}{18}\right), \qquad t \geq 0, \qquad P_{c(t)} = \psi^T M_t(u, v)\psi.$$

The full identity, before any state elimination, is

$$P_{c(t)} = P_{c(0)} + tD_u D_v X^2. \tag{61}$$

8.1 Positivity of the entire source family

For $uv \neq 0$, the upper WXY block of M_0 is positive definite. Indeed, all event packets have positive coefficients, and the restriction of the signed d-packet to $Z = 0$ is

$$\frac{1}{2}\left[(uW + uvX - Y)^2 + D_u v^2 Y^2 + D_v X^2\right].$$

The three unweighted rows $(u, uv, -1)$, $(0, 0, v)$, $(0, 1, 0)$ have determinant $-uv$. Thus their positive weighted Gram matrix is positive definite. With $U = u^2$, $V = v^2$, direct expansion gives

$$\det M_0 = D_u^2 D_v^2 UV\, L_0(U, V),$$

where

$$\begin{aligned} L_0(U, V) = \frac{121}{331776}\big(&880U + 9317U/V + 720V \\ &+ 880/V + 9477V/U + 720/U - 21994\big). \end{aligned} \tag{62}$$

This is nonnegative for $U, V > 0$ by weighted arithmetic–geometric mean. The six positive integer weights sum to 21994, and their weighted exponent vectors sum to $(0, 0)$:

$$880 + 9317 - 9477 - 720 = 0, \qquad -9317 + 720 - 880 + 9477 = 0.$$

Consequently the weighted geometric mean of $U, U/V, V, 1/V, V/U, 1/U$ is 1. The determinant is nonnegative, and the positive upper block makes its scalar Schur complement nonnegative. Hence $M_0 \succeq 0$ for $uv \neq 0$.

Continuity includes either angle equal to zero. Bihomogenizing the parent to degree two in each angle pair includes the projective endpoints by the same argument. Equivalently, signs are covered by the exact gauges

$$(u, v; W, X, Y, Z) \longmapsto (\epsilon u, \eta v; W, \eta X, \epsilon Y, \epsilon\eta Z), \qquad \epsilon, \eta \in \{-1, 1\}.$$

The additional term in (61) is nonnegative for $t \geq 0$, so the entire family is positive on all states and angles. A complex state gives the sum of the values on its real and imaginary parts; mixtures are handled by linearity. The same assertions hold over real closed fields by their first-order polynomial formulations.

8.2 The literal scalar remainders

Restrict to $v = 1$ and let $U = u^2 > 0$. Let A_t denote the upper 3×3 principal state block. Define

$$\begin{aligned} A(U) &= 26883U^2 + 105314U + 29403, \\ B(U) &= 29000U^2 + 50384U + 21384, \\ D_t(U) &= A(U) + tB(U), \\ N_t(U) &= 112167(U - 1)^2 + t(121000U^2 - 189648U + 81576). \end{aligned}$$

The determinant identities are

$$\det A_t = \frac{11}{20736}(U+1)D_t(U), \qquad \det M_t = \frac{11}{82944}(U+1)^2 N_t(U).$$

The literal Schur remainder after minimizing over W, X, Y with $Z = 1$ is therefore

$$S_t(U) = \frac{(U+1)N_t(U)}{4D_t(U)}. \tag{63}$$

Each coefficient of D_t is positive for $t \geq 0$, so these rational functions are defined for all $U > 0$.

The following exact identities control every possible cancellation:

$$\begin{aligned}
\operatorname{Res}_U(A, B) &= 1386911105418617856, \\
\operatorname{Res}_U(D_t, N_t) &= 10441728(8t+11)(1000t+927)(33677t+55550)^2, \\
D_t(-1) &= -49028, \\
\operatorname{disc}_U(D_t) &= 448(129472t^2 + 10942118t + 17699275).
\end{aligned}$$

In particular D_t is coprime to $(U+1)N_t$ for every $t \geq 0$. For distinct s, t, a common divisor of D_s, D_t would divide $(s-t)B$ and then A, contradicting the first resultant.

Proposition 8.1. *The functions $S_t : (0, \infty) \to \mathbb{R}$, $t \geq 0$, are linearly independent in the sense that every finite family with distinct parameters is linearly independent. They cannot all be sums of squares from a single fixed finite-dimensional space of functions of U.*

Proof. Suppose $\sum_{i=1}^{r} \lambda_i S_{t_i} = 0$ on $(0, \infty)$, with distinct t_i. Multiplication by $4\prod_i D_{t_i}$ yields a polynomial identity. Reduce it modulo D_{t_i}. Every term except the ith vanishes. Its remaining product consists of $(U+1)N_{t_i}$ and the other denominators, each coprime to D_{t_i}, so it is a unit in $\mathbb{R}[U]/(D_{t_i})$. It follows that $\lambda_i = 0$.

If V is a finite-dimensional function space, the linear span of all products of two elements of V is finite dimensional, of dimension at most $(\dim V)(\dim V + 1)/2$. Every sum of squares from V belongs to that product span. The established linear independence rules out a single such space for all S_t. □

8.3 Why the complete parent avoids this obstruction

The complete family already has five fixed testing features. Since M_0 is positive semidefinite, take the four entries of $M_0(u, v)^{1/2}\psi$ together with $\sqrt{D_u D_v}X$. The coefficients are $1, 1, 1, 1, t$, respectively. This is a fixed finite feature representation of (61), including arbitrary nonnegative source parameter t. The unique positive semidefinite square root can be chosen as a semialgebraic testing function.

Thus substituting a source-dependent state minimizer into a previously fixed feature space need not preserve that space. The literal-remainder argument cannot be used to reject the complete-parent local construction, nor does the five-feature example supply a global atlas for arbitrary quantum supports. The file `c2/schur_remainder_check.py` recomputes the complete parent, determinants, gauges, resultants and contact identities; its receipt records the precise finite checks.

9 Conditional effective algebraic extraction

9.1 What this note establishes

There are three separate conclusions. First, the membership formula below describes $\mathcal{Q}_{222}$ exactly. Second, real algebraic coefficient descent and a terminating search follow *if a finite lift of the searched type exists*. Third, the generated files contain one explicit candidate decision problem, not its solution. These distinctions are essential: a complete logical specification of an extraction problem does not supply the coefficient tuple requested by global extraction items E1–E3.

The finite-lift existence criterion of Scheiderer, Corollary 3.19 [4], requires a universal tensor sum-of-squares hypothesis. The criterion is not itself a proof that $\mathcal{Q}_{222}$ satisfies that hypothesis. The absent historical global proof is not reverified here. Likewise, existence of a semidefinite lift with arbitrary block order does not imply existence of a representation by blocks of order two. The second-order-cone search below is conditional on that stronger premise.

9.2 A rational atom map for the full quantum set

Use coordinates

$$q = (a_0, a_1, b_0, b_1, E_{00}, E_{01}, E_{10}, E_{11}).$$

One atom is parameterized by eight real numbers

$$z = (w, x, y, z_0, c_A, s_A, c_B, s_B)$$

satisfying

$$w^2 + x^2 + y^2 + z_0^2 = 1, \qquad c_A^2 + s_A^2 = 1, \qquad c_B^2 + s_B^2 = 1. \tag{64}$$

The state is the real vector (w, x, y, z_0) in the computational basis. Its observables are

$$A_0 = B_0 = Z, \qquad A_1 = c_A Z + s_A X, \qquad B_1 = c_B Z + s_B X.$$

Define the following quadratic polynomials:

$$\begin{aligned} T_A &= w^2 + x^2 - y^2 - z_0^2, & X_A &= 2(wy + xz_0), \\ T_B &= w^2 - x^2 + y^2 - z_0^2, & X_B &= 2(wx + yz_0), \\ C_{ZZ} &= w^2 - x^2 - y^2 + z_0^2, & C_{ZX} &= 2(wx - yz_0), \\ C_{XZ} &= 2(wy - xz_0), & C_{XX} &= 2(wz_0 + xy). \end{aligned}$$

The atom map Φ has coordinates

$$\begin{aligned} \Phi_1 &= T_A, & \Phi_2 &= c_A T_A + s_A X_A, \\ \Phi_3 &= T_B, & \Phi_4 &= c_B T_B + s_B X_B, \\ \Phi_5 &= C_{ZZ}, & \Phi_6 &= c_B C_{ZZ} + s_B C_{ZX}, \\ \Phi_7 &= c_A C_{ZZ} + s_A C_{XZ}, & & \\ \Phi_8 &= c_A c_B C_{ZZ} + c_A s_B C_{ZX} & & \\ &\quad + s_A c_B C_{XZ} + s_A s_B C_{XX}. & & \end{aligned} \tag{65}$$

Every coefficient is rational; the coordinate degrees are respectively

$$(2, 3, 2, 3, 2, 3, 3, 4).$$

The exact checker reconstructs all eight expressions independently from the corresponding 4×4 Pauli matrices and compares polynomial coefficients.

Theorem 9.1 (Nine-atom membership formula)**.** *Let Z denote the compact real algebraic set in (64). Then*

$$\mathcal{Q}_{222} = \operatorname{conv}(\Phi(Z)) = \left\{ \sum_{i=1}^{9} \lambda_i \Phi(z_i) : z_i \in Z,\ \lambda_i \geq 0,\ \sum_{i=1}^{9} \lambda_i = 1 \right\}.$$

This identity includes finite-dimensional complex and mixed states and arbitrary binary measurements. In particular, $\mathcal{Q}_{222}$ is compact, convex, and semialgebraic over $\mathbb{Q}$.

Proof. First dilate each binary observable, a self-adjoint contraction, to a reflection on a doubled local space. For an observable C the dilation

$$\widehat{C} = \begin{pmatrix} C & \sqrt{\mathbb{I} - C^2} \\ \sqrt{\mathbb{I} - C^2} & -C \end{pmatrix}$$

uses the same upper-block embedding for every choice of observable. It preserves the local expectations and correlators, so the dimension remains finite and all measurements may be taken projective.

Jordan's two-projector decomposition splits each local space into invariant subspaces of dimension at most two for its pair of reflections. Equivalently, one can diagonalize the unitary product of the reflections and pair conjugate eigenvalues; the resulting invariant blocks have dimension at most two. Applying the local decompositions to both parties makes every measured operator block diagonal. The trace against an arbitrary state is therefore a finite convex combination of expectations on pairs of local blocks. This is the standard qubit reduction underlying the extremal-correlation result of Masanes [2].

On a nontrivial two-dimensional block, a local unitary sends the first reflection to Z; a phase rotation preserving Z then makes the second reflection $cZ + sX$ with real c, s and $c^2 + s^2 = 1$. A one-dimensional block with reflection values $a_0, a_1 \in \{-1, 1\}$ embeds into a qubit by choosing the Z eigenvector of eigenvalue a_0 and using the second observable $a_0 a_1 Z$. Thus blocks of dimension one are included in the same parameterization.

All operators whose expectations enter the behavior are now real symmetric. If ρ is a complex density matrix on such a block pair, its entrywise real part is a real symmetric positive semidefinite matrix with trace one. Positivity follows first on real vectors from positivity of ρ, and then on complex vectors by writing a complex vector as its real and imaginary parts. The imaginary antisymmetric part of ρ has zero trace against every real symmetric measured operator, so replacing ρ by its real part preserves every coordinate. A real spectral decomposition then writes that state as a convex combination of real pure states. Each resulting behavior is an atom of (65).

Conversely, every atom is a physical two-qubit model, and every finite convex mixture is physical using a finite direct sum with a shared classical block label. Hence $\mathcal{Q}_{222} = \mathrm{conv}(\Phi(Z))$. Carathéodory's theorem in $\mathbb{R}^8$ reduces every finite mixture to at most nine atoms. Empty positions can be filled by arbitrary normalized atoms of zero weight. Since Z and the nine-weight simplex are compact, their continuous polynomial image is compact. The displayed finite existential description and real quantifier elimination imply semialgebraicity over $\mathbb{Q}$. Convexity follows from the same direct-sum mixture construction. □

9.2.1 Every quantified variable and constraint

The physical formula is

$$\begin{aligned} \mathrm{Phys}(q): \quad & \exists \lambda_1, \dots, \lambda_9 \; \exists z_1, \dots, z_9 : \\ & \lambda_i \geq 0, \quad \sum_{i=1}^{9} \lambda_i = 1, \quad z_i \text{ satisfies (64)}, \\ & q_\nu = \sum_{i=1}^{9} \lambda_i \Phi_\nu(z_i) \quad (\nu = 1, \dots, 8). \end{aligned} \tag{66}$$

The counts are exact:

Quantity	Count
Weights	9
Atom parameters	$9 \cdot 8 = 72$
Total existential variables	81
Weight-sum equation	1
State and circle equations	$9 \cdot 3 = 27$
Output coordinate equations	8
Total polynomial equations	36
Weight nonnegativity inequalities	9
Maximum total polynomial degree	5

The degree-five terms occur in $\lambda_i \Phi_8(z_i)$. No inequality selecting a measurement-angle chart is needed: the full circles are used. No square roots or trigonometric functions occur in (66). The formula is a membership description of the physical set, not a linear matrix inequality.

9.3 A fixed-pattern coefficient descent theorem

Fix positive block orders $r_1, \dots, r_M$ and let

$$V = \bigoplus_{j=1}^{M} \mathsf{S}^{r_j}, \qquad D = \dim V = \sum_{j=1}^{M} \frac{r_j(r_j+1)}{2}.$$

Here S^r is the real vector space of symmetric $r \times r$ matrices. A representation with this pattern has the form

$$K = \left\{ q \in \mathbb{R}^8 : \exists y \in \mathbb{R}^m, A_0 + \sum_{\nu=1}^{8} q_\nu A_\nu + \sum_{k=1}^{m} y_k B_k \in \prod_{j=1}^{M} \mathsf{S}_+^{r_j} \right\}. \tag{67}$$

Every coefficient is an element of V. The auxiliary term ranges over the linear span of $B_1, \dots, B_m$, whose dimension is at most D. Choosing a basis of that span and padding with zero directions if necessary yields the same represented set with exactly D allowed auxiliary variables. This step is linear algebra and does not require a bound from the historical existence proof.

Theorem 9.2 (Real algebraic coefficient descent)**.** *Let $K \subseteq \mathbb{R}^8$ be defined by a first-order formula in the ordered-ring language with rational coefficients. If K has a representation (67) with the fixed block pattern, it has such a representation with coefficients in the real algebraic numbers $\mathbb{R}_{\mathrm{alg}}$ and at most D auxiliary variables. For this fixed pattern, existence and validity of a specified algebraic coefficient tuple are decidable by real quantifier elimination.*

Proof. Use the preceding compression to allow D auxiliary variables. Let c be the finite tuple containing all independent coefficient entries. For fixed block orders, positive semidefiniteness is equivalent to nonnegativity of all principal minors. Therefore

$$\Theta(c): \quad \forall q \in \mathbb{R}^8 \left[\mathrm{Phys}_K(q) \iff \exists y \in \mathbb{R}^D : \; L(c; q, y) \succeq 0 \right] \tag{68}$$

is a first-order formula over $\mathbb{Q}$, where Phys_K is the given defining formula. Although an entry of L is affine in (q, y) when c is fixed, it is a polynomial in the joint variables (c, q, y), which is exactly what the ordered-ring language requires.

By the existence hypothesis, the rational sentence $\exists c\, \Theta(c)$ is true over $\mathbb{R}$. The real algebraic numbers form a real closed field, and real closed fields admit quantifier elimination. Consequently the sentence is true over $\mathbb{R}_{\mathrm{alg}}$ as well. Choose a witnessing tuple $c^* \in \mathbb{R}_{\mathrm{alg}}^n$. The inclusion $\mathbb{R}_{\mathrm{alg}} \subseteq \mathbb{R}$ is elementary; thus $\Theta(c^*)$ also holds over $\mathbb{R}$. This proves the claimed equality of represented sets with algebraic coefficients.

Effective quantifier elimination decides the sentence at the fixed pattern and decides $\Theta(c^*)$ when the algebraic numbers are given by exact real-root encodings. A sample point of the resulting

nonempty coefficient set can be computed by exact real algebraic algorithms. Alternatively, enumerate tuples of real algebraic numbers and decide Θ for each tuple. Once feasibility is known, coefficient descent guarantees that this enumeration eventually finds a witness. Both methods are effective in principle; neither is claimed to be practical for the input below. Background on the exact algorithms is given in [3]. □

The theorem does not assert the existence of rational coefficients. It also does not supply a bound on D, the block orders, the degrees of the algebraic coefficients, or the cost of the decision procedure.

9.4 Conditional search over all finite patterns

Suppose a full finite semidefinite lift of $\mathcal{Q}_{222}$ has been established independently. Enumerate a single positive semidefinite block of order $N = 1, 2, \ldots$. At each N, let $D = N(N+1)/2$, write the formula (68), and run an exact decision algorithm for $\exists c\, \Theta(c)$. Each fixed instance is decidable. Any finite product of positive semidefinite cones embeds into one block diagonal cone, so a positive decision is eventually reached under the finite-lift premise. One can then extract algebraic coefficients and verify both inclusions by deciding Θ with those coefficients substituted.

If the stronger premise is a lift by finitely many blocks of order at most two, enumerate the number M of 2×2 blocks instead. Scalar positive semidefinite blocks can be padded by a zero second row and column. The same conditional termination argument applies. Mere existence of an arbitrary finite semidefinite lift supplies no termination guarantee for the order-two-only enumeration.

For the full-dimensional quantum set, an arbitrary affine projection representation can be put into the coordinate form (67): the projection has rank eight, so complete its eight output coordinates to affine coordinates on its parameter space. Full affine dimension can already be seen in the 16 deterministic local behaviors

$$(a_0, a_1, b_0, b_1, a_0b_0, a_0b_1, a_1b_0, a_1b_1), \qquad a_0, a_1, b_0, b_1 \in \{-1, 1\}.$$

The regenerated rational row reduction gives affine rank eight. This is independent of any unverified global lift claim.

9.5 The three-block input: exact specification, unsolved status

For M blocks of order two, the matrix-space dimension is $D = 3M$. Use D auxiliary variables, and give each of the three independent entries in each block an arbitrary affine coefficient for the constant, the eight quantum coordinates, and the D auxiliary coordinates. The number of coefficient unknowns is

$$3M(1 + 8 + 3M) = 3M(9 + 3M).$$

For $M = 3$ this gives nine auxiliaries and 162 coefficient unknowns. For block j, write

$$L_j(c; q, y) = \begin{pmatrix} a_j(c; q, y) & b_j(c; q, y) \\ b_j(c; q, y) & d_j(c; q, y) \end{pmatrix}.$$

Its exact positivity formula is

$$a_j \geq 0, \qquad d_j \geq 0, \qquad a_j d_j - b_j^2 \geq 0.$$

All diagonal conditions are retained, including singular and zero blocks. The candidate sentence is

$$\exists c \in \mathbb{R}^{162}\ \forall q \in \mathbb{R}^8 : \qquad \mathrm{Phys}(q) \quad \longleftrightarrow \quad \exists y \in \mathbb{R}^9 :\ L_1, L_2, L_3 \succeq 0. \tag{69}$$

The symbol $\mathrm{Phys}(q)$ in (69) is the complete 81-variable existential formula (66), not an oracle or an approximate quantum membership test.

The value $M = 3$ is the first not excluded by a simple dimension obstruction in this direct-coordinate format. Indeed, if the linear map $q \mapsto \sum q_\nu A_\nu$ had a nonzero kernel direction v, every feasible q would be accompanied by the whole line $q + \mathbb{R}v$, with the same auxiliary witness. Since $\mathcal{Q}_{222}$ is nonempty and bounded, that map must be injective. Hence $3M \geq 8$, which excludes $M = 1, 2$. This necessary condition does not imply feasibility at $M = 3$; no such inference is made.

The Wolfram Language file constructs the fully expanded rational formulas and the sentence, with all symbols explicitly named. It does not invoke `Resolve`, `Reduce`, `FindInstance`, or any optimization routine. The companion JSON contains the exact sparse polynomial terms, variable order, coefficient mapping, counts and status. Both are labelled `UNSOLVED`. No global coefficient tuple is present.

9.6 Fresh encoding checks and the remaining extraction gates

Run `python3 generate_q222_qe_problem.py`. It uses only the shared standard-library exact polynomial helper. Its receipt records 20 fresh encoding checks: all eight Pauli atom coordinates, the atom degree vector, the membership counts and degree bounds, rational affine rank of the deterministic behaviors, coefficient and auxiliary counts, disjoint symbol names, and the nine scalar positive-semidefinite conditions. These checks establish the advertised input encoding. They do not establish candidate feasibility, run quantifier elimination, or provide algebraic lift coefficients.

The remaining global extraction gates are precise:

1. Establish and retain a complete verified full-set existence premise, or independently find a valid finite block pattern and its coefficients.

2. Evaluate a global finite feature system or solve a candidate formula such as (69); fix the actual matrix order and auxiliary dimension.

3. Export every coefficient in the original eight coordinates using exact real algebraic encodings, including root-selection data.

4. Prove both full-set inclusions with those fixed coefficients. A proposed lift can be checked by deciding (68), or by independent exact analytical certificates for each direction.

5. Replay the coefficient verification, bind the evidence and data by hashes, and mark E1–E3 complete only after their actual deliverables exist.

The present note completes the physical membership encoding and the conditional logical route. It leaves these full-set extraction gates open. The separately regenerated tilted–CHSH projection has evaluated coefficients and a proof of both inclusions for that projection, but its explicit nonquantum scope witness prevents using those coefficients as a full-set solution.

10 Extraction checklist and remaining data

The following is the supplementary checklist for this regenerated refinement. It does not recertify the original R21 existence checklist. The status of each item is limited to its stated domain. Thirteen items are complete within those scopes and seven remain open. Global items E1–E3 are open.

ID	Status	Acceptance and result
S01	COMPLETE	**Recover and bind the retained source inputs**. The canonical August 30 source and the August 31 frozen ZIP match their recorded SHA-256 digests. The earlier continuation archive is preserved unchanged. This verifies custody, not all inherited mathematics.

ID	Status	Acceptance and result
L01	COMPLETE	**Explicitly separate every positive local coefficient**. The finite identity gamma=p(x)+q(y)+sum(A_i(x)+delta)(B_i(y)+delta) has explicit strict sign bounds and a fixed order-two block conversion. The toy 1+xy example is fully evaluated.
L02	COMPLETE	**Prove the denominator-free smooth-diagonal construction**. Given exact J^2 data and positive tangential residue, add a normal Gram term, use a constant matrix change, and apply the fixed m^2-frame identity. The theorem leaves its global hypotheses to be instantiated.
L03	COMPLETE LOCAL	**Evaluate a full regular-C2 contact box**. The complete parent, six contact equations, regular contact graph, exact Taylor matrix, 25 frames and rational radius are evaluated. The source/contact box and testing box vary independently. This is one normalized local domain.
L04	COMPLETE LOCAL	**Evaluate the regular-C2 feature list and product map**. Delivered: 785 order-two blocks, 1570 listed feature entries with repetitions, 237 ambient monomials, 27747 product coefficients, and 295 target coefficients in seven source columns. These do not specify a global eight-coordinate quantum lift.
B01	COMPLETE MODULE	**Extract the displayed B0 coefficient module**. The four displayed coefficient scalars form exactly the nonnegative orthant, with four rational necessity tests, nine polynomial square features, and an order-four diagonal pencil. August repair R5 factor typing is retained.
B02	COMPLETE CONDITIONAL	**Bound the feature envelope under a fixed-testing-variable congruence**. Twenty monomials suffice when the 4-by-4 state congruence is independent of polynomial-testing variables U,V. This is a conditional envelope; it does not assert that order-two blocks survive or that the physical restoration map is known.
B03	OPEN	**Supply the actual B0 radial restoration and clearing maps**. Required: the complete state congruence, every pure-factor scaling/clearing function and its domain, the original slack identity, original-coordinate map, and every singular or zero-factor branch. The retained base asserts a congruence but does not print these data.
C01	COMPLETE RESTRICTED	**Reconstruct the restricted event-contact cofactor vector**. All six reduced forms, thirty Jacobian entries and six primitive cofactors are evaluated; all five JX=0 identities hold. This is a restricted stationary-contact parametrization, not a proof of global physical validity or boundary completeness.
T01	COMPLETE	**Certify the complete tilted-CHSH support family**. The reflection SOS, attainment, all real tilt parameters, both signs, endpoint completion, and separately labelled real-closed-field factors are covered for the stated two-coordinate projection.
T02	COMPLETE	**Evaluate the tilted fixed features and affine pencils**. The order-five lift has one auxiliary and 14 nonzero upper coefficient records over Q(sqrt(2)); elimination gives order four, zero auxiliaries and 12 records. Original-coordinate substitution gives a projection cylinder.
T03	COMPLETE	**Prove both projected inclusions and exhibit the full-set limitation**. Both inclusions are proved for |t|<=1 and t^2+s^2/4<=2. The rational point (0,0,0,0,1,1,1/4,-1/4) passes all probability and relabelled tilted-family inequalities but is nonquantum.

ID	Status	Acceptance and result
D01	COMPLETE	**Audit the literal Schur-remainder extraction route**. A physical one-parameter complete-parent family has pairwise coprime literal remainder denominators and infinitely many linearly independent scalar remainders. This rules out that literal fixed-feature route, while the complete family itself has five fixed testing features.
A01	OPEN	**Evaluate a finite global all-source/all-test atlas**. Required: a finite cover of the whole product domain, including exceptional, zero, boundary, projective, away-from-contact and normalization branches, with every restored slack identity. One regular rectangle and general local theorems do not establish the cover.
G1	OPEN	**Settle the retained enlarged-candidate C1 admission gate**. For the retained order-13744 candidate route, the stated C1 canonical admission/converse gate remains open. An alternative exact full-set construction could bypass this sufficient route. It is not made true by the local C2 extraction.
G2	OPEN	**Settle the retained enlarged-candidate C2 admission gate**. For the same retained enlarged candidate, the universal C2 admission/converse gate remains open. Exactness on the reciprocity-fixed sector or near one contact does not prove this universal gate.
E1	OPEN	**Deliver the evaluated global feature list and actual dimensions**. Required: one fixed finite global list with all radicands, domains, auxiliary variables and a proved completeness statement for every valid full-set affine slack. The present local counts are not global bounds.
E2	OPEN	**Deliver the evaluated full eight-coordinate affine pencil**. Required: actual finite N,m and every coefficient of A0 + sum(q_nu A_nu) + sum(y_j B_j), exact algebraic encodings where necessary, and the projection to the original eight coordinates. No such exact full-set pencil is delivered.
E3	OPEN	**Prove both full-set inclusions for that fixed pencil**. Required: physical-point lifting and exclusion of every nonquantum point using the delivered fixed coefficients, with exact replayable coefficient evidence. A projection-only proof, an outer approximation or unsolved QE formula does not satisfy this gate.
Q01	COMPLETE CONDITIONAL	**Specify a conditional exact algebraic extraction procedure**. The full rational nine-atom physical formula, real-algebraic coefficient descent theorem and fixed-pattern decision input are explicit. The M=3 order-two input is UNSOLVED: feasibility is not established and no quantifier elimination was run. Termination of the pattern search requires the corresponding existence premise.

10.1 Data that would complete the global extraction

The labels G01–G10 below distinguish these concrete data obligations from the retained enlarged-candidate gates G1 and G2. All ten obligations remain open as global requirements. Local pieces delivered here may satisfy parts of them.

G01: Chart inventory. List every physical source and testing chart with its original-coordinate map, normalization and domain. Include all signs, axes and projective endpoints.

G02: Full-slack restoration. Print and verify every exact state/observable congruence and scaling from a displayed local normal form back to the complete original slack. Handle every vanishing clearing factor.

G03: Regular contacts. Evaluate sufficient regular anchors and their exact local matrices, radii, feature lists and source/test domains. Prove the asserted portion of the regular contact locus is covered.

G04: Exceptional and boundary contacts. Supply exact certificates for rank drops, weak and zero openings, event boundaries, state axes and projective limits. Do not divide by a parameter allowed to vanish.

G05: Away-from-contact region. Supply fixed separated positivity certificates on the remaining source/test cells with explicit valid domains and normalization control.

G06: Finite global coverage. Prove the finite rectangles or finite cell atlas cover the complete product of all normalized valid supports and all physical tests. Deal explicitly with noncompact parameter directions.

G07: Global feature system. Apply the finite cell construction, restore affine constants and homogeneities, and publish every fixed function with its radicand/domain and block membership. Determine actual dimensions.

G08: Global product coefficients. Evaluate a basis containing 1 and the original eight coordinate functions plus all within-block products. Publish every exact expansion and any algebraic root-selection data.

G09: Original-coordinate pencil. Write the actual matrices A0,A1,...,A8,B1,...,Bm and all equality constraints or their scalar-block encodings. Record index, symmetry, trace and projection conventions.

G10: Both global inclusions and release. Prove physical points lift and every feasible projection is physical for the same fixed coefficients. Replay exact checks, bind source/dependency evidence, inspect the PDF and verify the complete archive.

10.2 Where the retained outer candidate stands

The unchanged earlier continuation archive records an evaluated outer candidate with matrix order 13744, 29920 auxiliary variables, 2928 blocks and 153248 upper coefficient records. Its recorded global reverse inclusion remains unproved. These figures and the earlier rejected orders 8368 and 10736 are inherited context; the old arrays are preserved without a new full replay in this regeneration. Neither local extraction nor the general assembly theorem upgrades that candidate to an exact full-set lift.

10.3 Reading the verification records

The root verification report identifies each fresh replay and its scope. Hash checks establish file integrity and custody; exact algebra checkers corroborate the specific identities they evaluate. Their overlapping assertion counts are not added into a mathematical completeness total. The analytical and separate-agent reviews are within the same model system. They are neither external peer review nor proof-assistant formal verification.

References

[1] C. Bamps and S. Pironio, *Sum-of-squares decompositions for a family of CHSH-like inequalities and their application to self-testing*, Physical Review A **91**, 052111 (2015), arXiv:1504.06960. The operator SOS used here is equation (27) in arXiv version 1.

[2] Ll. Masanes, *Extremal quantum correlations for N parties with two dichotomic observables per site*, arXiv:quant-ph/0512100 (2005).

[3] S. Basu, *Algorithms in Real Algebraic Geometry: A Survey*, arXiv:1409.1534 (2014), especially Section 2.

[4] C. Scheiderer, *Second-order cone representation for convex subsets of the plane*, arXiv:2004.04196, version 2 (2021), Corollary 3.19.